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Fundamentals of Data Science

Unit-1

Pattabhirama Mohan

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Introduction to Data Science

Pattabhirama Mohan

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Extracting Meaningful Insights from Raw Information

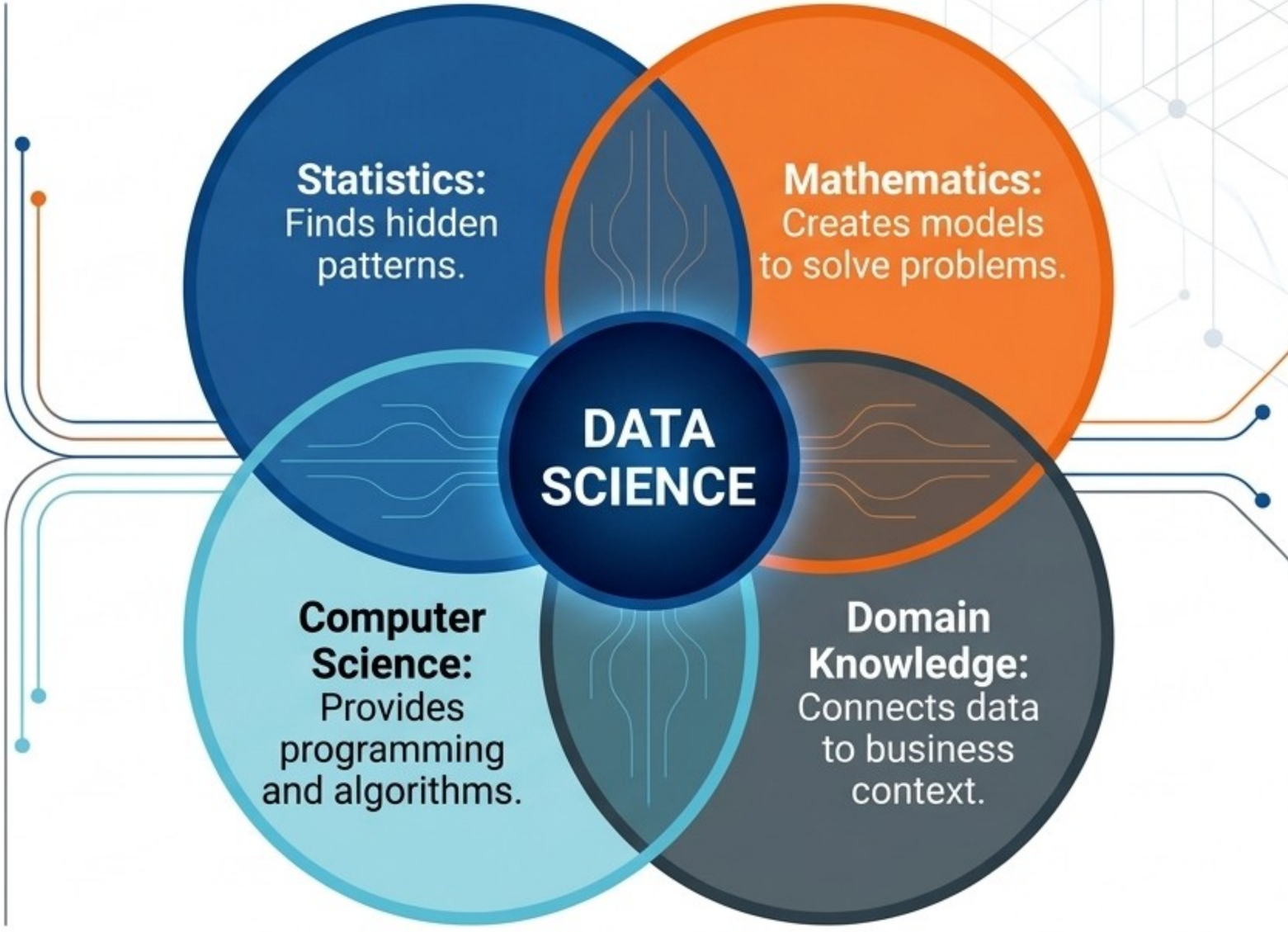
What is Data?

Data is raw information categorized as:

Structured: Excel sheets, relational databases (Numbers).

Unstructured: Emails, social media, videos (Images/Text).

Semi-structured: XML, JSON files.



NotebookLM

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SCOPE OF DATA SCIENCE

Data Science helps turn data into valuable insights and supports decision-making in almost every industry.

1. HEALTHCARE

Applications
- Disease prediction
- Medical image analysis
- Patient monitoring
- Drug discovery

Example: Predicting heart disease risk using patient health records.

2. FINANCE

Applications
- Fraud detection
- Credit scoring
- Stock market prediction
- Risk analysis

Example: Banks detect fraudulent transactions in real-time.

3. E-COMMERCE / RETAIL

Applications
- Product recommendation
- Customer segmentation
- Demand forecasting
- Price optimization

Example: Amazon recommends products based on previous purchases.

4. EDUCATION

Applications
- Student performance prediction
- Personalized learning
- Attendance analytics
- Course recommendation

Example: Predicting student performance to provide extra help early.

5. AGRICULTURE

Applications
- Crop yield prediction
- Soil analysis
- Weather forecasting
- Smart irrigation

Example: Predicting crop yield based on soil and weather data.

DATA SCIENCE

Extracting Knowledge and Insights from Data for Better Decisions

6. MANUFACTURING

Applications
- Predictive maintenance
- Quality inspection
- Production planning
- Supply chain optimization

Example: Predicting machine failure before it happens to reduce downtime.

7. TRANSPORTATION

Applications
- Traffic prediction
- Route optimization
- Fleet management
- Autonomous vehicles

Example: Optimizing delivery routes to save fuel and time.

8. SOCIAL MEDIA & MARKETING

Applications
- Sentiment analysis
- Trend detection
- Customer behavior analysis
- Targeted advertising

Example: Analyzing tweets to understand public opinion about a brand.

9. GOVERNMENT & PUBLIC SERVICES

Applications
- Policy planning
- Crime prediction
- Resource allocation
- Smart city development

Example: Analyzing crime data to deploy resources in high-risk areas.

10. ENTERTAINMENT & MEDIA

Applications
- Audience behavior analysis
- Movie / content recommendation
- Box office prediction
- Ad effectiveness analysis

Example: Netflix recommends shows based on viewing history.

In Short: Data Science is everywhere – it helps organizations make smarter decisions, improve efficiency, reduce costs, and create better experiences.

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HISTORY AND EVOLUTION OF DATA SCIENCE

From Data to Decisions: The Journey of Data Science

1960s

Statistical Foundations

- Data analysis began with statistics and mathematical methods.
- Focus on inference, surveys, and hypothesis testing.



Key Scientist

John Tukey
Pioneer of Exploratory Data Analysis.



1970s

Data Management Begins

- Databases and file systems emerged.
- Data stored, organized and retrieved efficiently.



Key Scientists



Edgar F. Codd
Father of Relational Database Model.

1980s

Rise of Data Warehousing

- Organizations started storing large volumes of integrated data.
- Data warehouses support business reporting and decision making.



Key Scientist



Bill Inmon
Father of Data Warehousing.

1990s

Emergence of Data Mining

- New techniques developed to extract patterns and discover knowledge.
- Growth of machine learning algorithms.



Key Scientists



Jiawei Han
Pioneer in Data Mining.



Gregory Piatetsky-Shapiro
Known as the Father of KDD.

2000s

The Term "Data Science"

- "Data Science" term was coined.
- Academic programs and industry roles started emerging.



Key Scientist



D.J. Patil
Co-founder of Kaggle. Advocate for Data Science as a discipline.

2010s

Big Data Revolution

- Explosion of data from social media, sensors, and mobile devices.
- Big data technologies like Hadoop, Spark became popular.



Key Scientist



Jeff Dean
Google Senior Fellow. Key contributor to Big Data systems (MapReduce).

2020s and Beyond

AI-Powered Data Science

- Deep learning, generative AI, and real-time analytics transform industries.
- Data science becomes more automated, intelligent and accessible.



Key Scientist



Fei-Fei Li
Leader in AI and Human-Centered AI research.

	1960s	1970s	1980s	1990s	2000s	2010s	2020s and Beyond
Key Developments	Statistical methods, surveys, hypothesis testing, sampling techniques.	Relational databases, DBMS, data models, network databases.	Data Warehouses, ETL (Extract, Transform, Load), OLAP.	Data Mining techniques (Association, Clustering, Classification, Prediction).	Statistical learning, Machine learning beginnings, R and Python rise.	Big Data technologies (Hadoop, Spark), NoSQL databases, Cloud computing.	Deep Learning, Generative AI (GPT, DALL·E), Real-time analytics, Edge AI, AutoML.
Real-world Example	Use of statistics in agriculture experiments, quality control.	Banking systems using relational databases to store customer data.	Retail companies building data warehouses for sales reporting.	Market basket analysis in retail stores.	Google uses data science for search ranking and ad targeting.	Facebook analyzes user data for personalized experiences.	ChatGPT, self-driving cars, fraud detection in real time.

KEY ENABLERS OF EVOLUTION

- Advances in Computer Hardware
- Growth of Internet
- Cloud Computing
- Big Data Technologies
- Machine Learning & AI Advancements
- IoT and Mobile Devices
- High Performance Computing

OUTCOME

Data Science has evolved from simple statistical analysis to a powerful interdisciplinary field that combines statistics, computer science, domain knowledge, and AI to solve complex real-world problems and drive data-informed decisions across all industries.

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FOUNDATIONS OF DATA SCIENCE

Data Science combines multiple disciplines to extract meaningful insights from raw data and solve real-world problems.

1. STATISTICS

Finds hidden patterns in data.

- Collecting & organizing data
- Summarizing & describing data
- Identifying relationships
- Hypothesis testing
- Making predictions

Techniques

Mean, Median, Mode, Regression, Correlation, Probability, Hypothesis Testing, etc.

EXAMPLE

A hospital analyzes patient data and finds that high blood pressure is a major risk factor for heart disease.



2. MATHEMATICS

Creates models to solve problems.

- Linear Algebra
- Calculus
- Probability
- Optimization
- Discrete Mathematics

Role

Builds algorithms, models, and optimization techniques for accurate predictions.

EXAMPLE

Netflix uses mathematical models to recommend movies based on user preferences and behavior.



DATA SCIENCE

Extracting Meaningful Insights from Raw Data

3. COMPUTER SCIENCE

Provides tools, programming and algorithms to process large volumes of data.

- Programming (Python, R, SQL)
- Data Structures & Algorithms
- Databases
- Big Data Technologies
- Cloud Computing
- Machine Learning

Role

Collects, cleans, stores, processes data and builds models at scale.

EXAMPLE

Amazon processes millions of transactions daily using distributed systems and recommends products in real time.



4. DOMAIN KNOWLEDGE

Connects data with real-world business or application context.

- Understands business problems
- Selects relevant data
- Interprets model outputs
- Supports decision making
- Adds practical perspective

Role

Ensures insights are relevant, meaningful and actionable.

EXAMPLE

A doctor interprets a model's prediction about diabetes risk and decides the right treatment for the patient.



HOW THE FOUNDATIONS WORK TOGETHER

Raw Data
Structured, Semi-structured or Unstructured

→

Collect & Process
Data collection, cleaning, transformation

→

Analyze & Model
Statistical analysis and mathematical modeling

→

Implement
Algorithms, programming and computation

→

Interpret & Apply
Domain expertise to derive insights and decisions

→

Outcomes
Better decisions, predictions, innovation and impact

KEY TAKEAWAY

Data Science is an interdisciplinary field. When Statistics, Mathematics, Computer Science and Domain Knowledge work together, they transform raw data into valuable insights that drive better decisions and create real-world impact.

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DISTINGUISHING DATA SCIENCE, AI AND MACHINE LEARNING

1. DATA SCIENCE (DS)



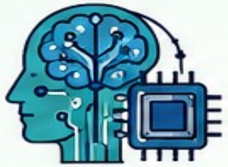
An interdisciplinary field that extracts meaningful insights from data to support decision-making.

PRIMARY GOAL	Analyze data and support decision-making.
FOCUS	Data collection, cleaning, analysis, visualization and interpretation.
OUTPUT	Reports, dashboards, business insights and recommendations.
SCOPE	Broadest field covering the complete data lifecycle.
KEY TOOLS	Python, SQL, Tableau, R

EXAMPLE

E-commerce company analyzes sales data to find best-selling products, customer trends and improve business strategy.

2. ARTIFICIAL INTELLIGENCE (AI)



A branch of computer science that develops systems capable of performing tasks that require human intelligence.

PRIMARY GOAL	Build intelligent systems that can think, reason and act.
FOCUS	Human-like intelligence, reasoning, planning, perception and automation.
OUTPUT	Intelligent applications such as chatbots, robots, assistants and automation.
SCOPE	Broad field that includes ML, robotics, expert systems, NLP, computer vision, etc.
KEY TOOLS	Expert Systems, NLP Frameworks, Neural Networks, Robotics

EXAMPLE

A chatbot understands customer questions and responds automatically like a human assistant.

3. MACHINE LEARNING (ML)



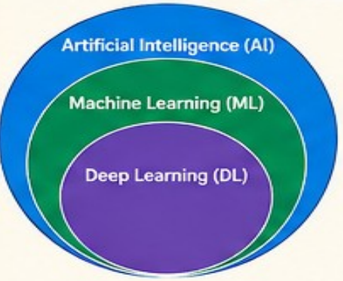
A subset of AI that enables computers to learn from data without being explicitly programmed.

PRIMARY GOAL	Develop models that learn patterns from data and make predictions.
FOCUS	Learning algorithms, prediction, classification and improvement.
OUTPUT	Predictive models and trained algorithms.
SCOPE	Narrower field focused on learning from data.
KEY TOOLS	Scikit-Learn, TensorFlow, PyTorch

EXAMPLE

Netflix recommends movies to users based on their watch history using machine learning models.

RELATIONSHIP AMONG DS, AI AND ML



- AI is the largest field that aims to create intelligent systems.
- ML is a subset of AI that enables systems to learn from data.
- Deep Learning (DL) is a subset of ML that uses neural networks with many layers.
- Data Science uses statistics, programming, visualization and ML to extract insights and support decisions.

HOW THEY WORK TOGETHER



DATA	PROCESS	ANALYZE	INSIGHTS	DECISIONS
Collect data from various sources	Clean, prepare and organize data	Use statistics, ML algorithms and AI	Generate meaningful insights	Make better data-driven decisions

QUICK COMPARISON

Feature	Data Science	AI	ML
Purpose	Extract insights from data	Create intelligent systems	Learn from data
Main Focus	Data analysis	Human-like intelligence	Learning and prediction
Uses Data?	✔	✔	✔
Creates Models?	Sometimes	Yes	Yes
Produces Reports?	Yes	No	No
Example	Sales Dashboard	Chatbot	Recommendation System

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THE FOUR CORE BUILDING BLOCKS OF DATA ANALYSIS

Transforming Raw Data into Meaningful Insights

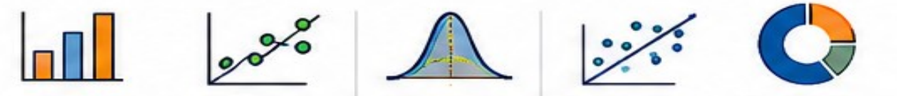
DATA SCIENCE CORE

1. DATA ARCHITECTURE



Role: The systems, pipelines, and structures that store, organize and manage data securely and efficiently.

2. STATISTICS & PROBABILITY



Role: The mathematical foundation used to analyze data, validate patterns and prove that results are meaningful.

3. BIG DATA ENGINE



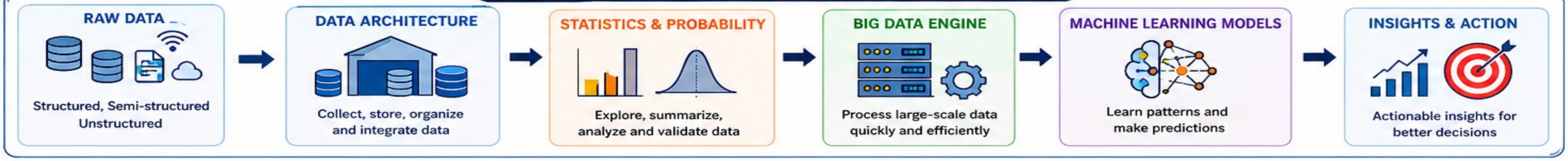
Role: The infrastructure to handle massive, fast-moving datasets that traditional databases cannot process.

4. MACHINE LEARNING MODELS



Role: The algorithms that learn from historical data to make accurate predictions and decisions.

HOW THESE FOUR BUILDING BLOCKS WORK TOGETHER



BENEFITS

- Better Understanding of Data
- Accurate Predictions and Forecasts
- Operational Efficiency and Automation
- Data-Driven Decision Making
- Competitive Advantage and Innovation

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A STRUCTURED APPROACH TO SOLVING DATA PROBLEMS

Data Science Life Cycle

DATA SCIENCE LIFE CYCLE

1 BUSINESS UNDERSTANDING

- Define the problem.
- Understand objectives.
- Identify success criteria.

Example:
Predict students who are at risk of failing.

2 DATA ACQUISITION

- Collect data from various sources.
- Databases, APIs, files, web, sensors.
- Ensure data is relevant.

Example:
Collect attendance, marks, demographics from university database.

3 DATA PREPARATION

- Clean the data.
- Handle missing values and duplicates.
- Transform and format data.

Example:
Fill missing marks, remove duplicates, standardize formats.

4 EXPLORATORY DATA ANALYSIS (EDA)

- Summarize data using statistics.
- Visualize to find patterns and trends.
- Identify outliers and relationships.

Example:
Visualization shows low attendance students score low in exams.

5 MODELING

- Apply Machine Learning algorithms.
- Train the model on historical data.
- Evaluate and tune the model.

Example:
Build a model to predict at-risk students.

6 DEPLOYMENT & COMMUNICATION

- Deploy the model into production.
- Create dashboards and reports.
- Communicate insights to stakeholders.

Example:
Dashboard alerts faculty about at-risk students for timely support.

EXAMPLE: STUDENT PERFORMANCE PREDICTION

1. **Define Problem** — Predict students at risk of failing.
2. **Collect Data** — Attendance, marks, assignments, exams.
3. **Clean Data** — Handle missing values, remove errors.
4. **Analyze Data** — Find patterns & trends using EDA.
5. **Build Model** — Train ML model to predict performance.
6. **Deploy & Share** — Alert faculty and support students on dashboard.

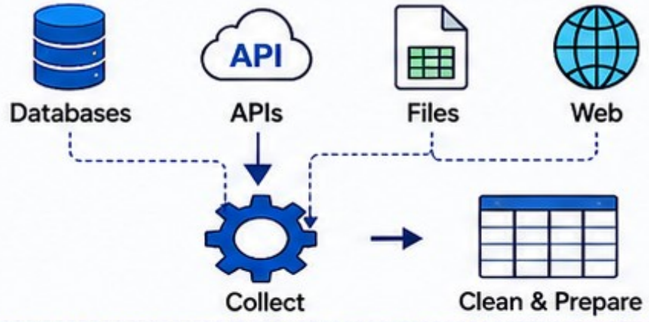
Better Decisions — Improve results and student success.

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MATCHING THE RIGHT TECHNOLOGY TO THE TASK

The Data Science Technology Stack

1 DATA COLLECTION & PREPARATION



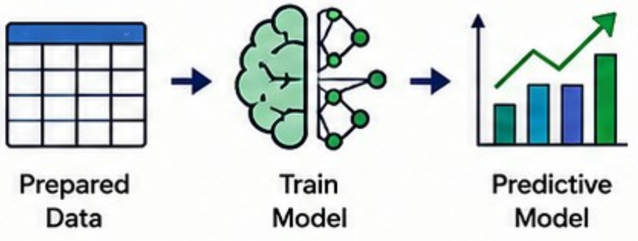
TECHNOLOGIES USED

Languages: Python, R, SQL

Libraries (Python): Pandas, NumPy

Action: Extracting, cleaning and organizing raw data.

2 MACHINE LEARNING & MODELING

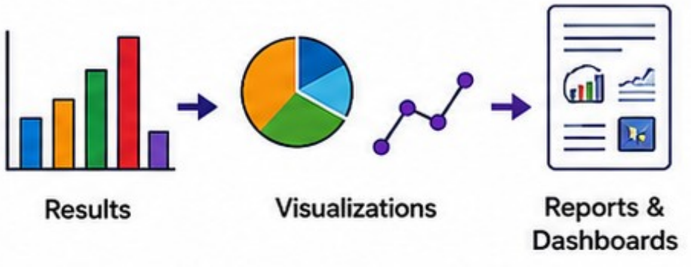


TECHNOLOGIES USED

Frameworks: Scikit-Learn, TensorFlow, PyTorch

Action: Training algorithms to learn patterns and make predictions.

3 VISUALIZATION & REPORTING



TECHNOLOGIES USED

Software / Tools: Tableau, Power BI, Matplotlib

Action: Turning complex numbers and results into easy-to-read charts and dashboards.

EXAMPLE: STUDENT PERFORMANCE PREDICTION WORKFLOW

1. **Collect Data** — Attendance, marks, assignments from university database.
2. **Clean & Prepare** — Handle missing values, remove duplicates, standardize data.
3. **Train Model** — Use ML algorithms to predict at-risk students.
4. **Analyze Results** — Evaluate model performance and identify key factors.
5. **Visualize & Report** — Create dashboards for faculty and administrators.

Better Decisions — Provide support to at-risk students and improve outcomes.

BENEFITS

- Right data for better accuracy
- Faster analysis and predictions
- Cost and time efficiency
- Better decision making
- Improved performance and growth

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TRANSFORMING INDUSTRIES BY TURNING GUESSES INTO FACTS

Data Science Applications Across Industries

1 BANKING & FINANCE

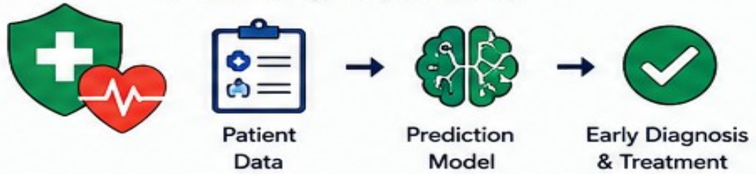
Detecting fraudulent transactions in real-time.



Benefits: Prevents fraud, reduces losses, improves customer trust.

2 HEALTHCARE

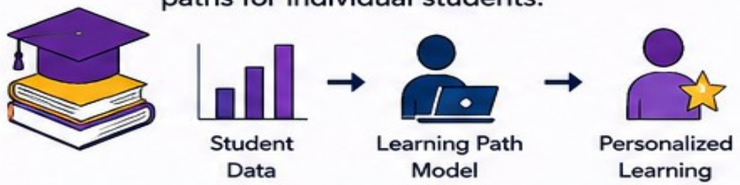
Predicting patient diseases and accelerating drug discovery.



Benefits: Early detection, better treatment, saves lives.

3 EDUCATION

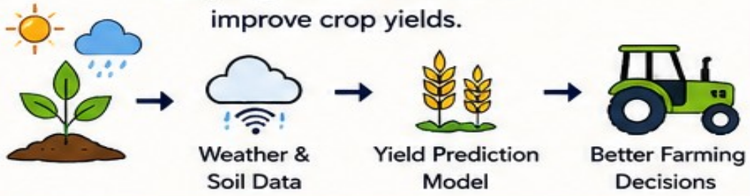
Creating customized learning paths for individual students.



Benefits: Improved learning outcomes, better student engagement.

4 AGRICULTURE

Analyzing weather and soil to improve crop yields.



Benefits: Higher productivity, optimized resources, reduced costs.

5 E-COMMERCE

Recommending products based on browsing history.



Benefits: Better user experience, increased sales and customer loyalty.

6 MANUFACTURING

Predicting when machinery will break down via Predictive Maintenance.



Benefits: Reduced downtime, lower maintenance costs, higher efficiency.

THE DATA SCIENCE PROCESS (Common to All Industries)



1. Collect Data
Gather data from various sources.

2. Prepare Data
Clean, transform and organize the data.

3. Explore Data
Analyze and discover patterns & insights.

4. Build Models
Train machine learning models.

5. Evaluate
Test and evaluate model performance.

6. Deploy
Deploy the model for real-world use.

7. Decide & Improve
Make data-driven decisions and continuously improve.

From Data to Decisions → Creating Value in Every Industry

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POWERING E-COMMERCE AND DIGITAL ENTERTAINMENT

How Data Science Personalizes Your Experience

1. AMAZON: THE RECOMMENDATION ENGINE

Recommends the right products based on your shopping behavior and millions of similar customers.

1. DATA COLLECTION

- Past purchases
- Items in your cart
- Products viewed
- Search history
- Similar customers' purchases

2. DATA ANALYSIS

Machine Learning finds patterns and relationships from millions of transactions.

3. RECOMMENDATION

Predicts products you are most likely to buy next.

4. RESULT

Personalized product recommendations for you!

EXAMPLE

You search for a Laptop

Amazon Recommends:

Laptop Bag ★★★★½

Wireless Mouse ★★★★½

USB Hub ★★★★½

Headphones ★★★★½

BUSINESS IMPACT

35% of Amazon's total revenue comes from personalized recommendations.

2. NETFLIX: CONTENT PERSONALIZATION

Recommends the right movies and shows based on your viewing behavior and preferences.

1. DATA COLLECTION

- What you watch
- When you pause
- What you skip
- Genres you prefer
- Ratings & searches

2. MACHINE LEARNING

Algorithms analyze your interests and find similar users with same taste.

3. PERSONALIZED PICKS

Ranks thousands of titles and selects the best ones for you.

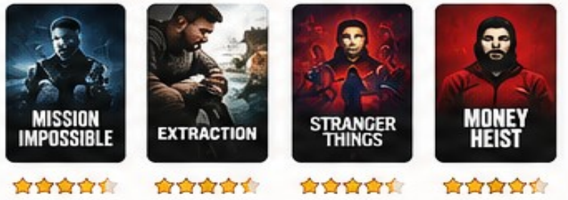
4. RESULT

Movies & shows you are most likely to enjoy!

EXAMPLE

You watch Action & Thriller movies

Netflix Recommends:



BUSINESS IMPACT

Keeps users engaged and significantly reduces subscription cancellations.

KEY COMPARISON

Aspect	Amazon	Netflix
Goal	Recommend the right products	Recommend the right movies & shows
Data Used	Purchases, searches, cart, views, ratings	Watch history, pauses, skips, ratings, searches
Technology	Machine Learning, Recommendation Algorithms	Machine Learning, Recommendation Algorithms
Outcome	Increased sales, better customer experience	Higher engagement, customer retention
Example	You search a laptop → recommends laptop bag	You watch thrillers → recommends laptop movies

HOW IT WORKS: THE RECOMMENDATION PIPELINE

Collect User Data
From multiple sources

→

Analyze Patterns
Find relationships and similarities

→

Build ML Models
Train algorithms on big data

→

Generate Recommendations
Predict what user will like

→

Deliver Personalized Experience
Show the right products / content

BENEFITS FOR USERS & BUSINESSES

- Personalized Experience
- Saves Time & Effort
- Increases Sales & Revenue
- Improves Customer Satisfaction
- Increases User Engagement
- Builds Loyalty & Trust

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OPTIMIZING NAVIGATION AND ADVANCING HEALTHCARE

Data Science Making Life Better

1. GOOGLE MAPS: ROUTE OPTIMIZATION

Finding the fastest, safest, and most efficient routes in real-time.

1. DATA COLLECTION



- Live location from millions of devices
- Traffic speed
- Road conditions
- Accidents & closures
- Weather updates

2. DATA ANALYSIS



Algorithms analyze speed, traffic density, historical data and road conditions.

3. PREDICTION



Predicts travel time, traffic congestion and best possible routes.

4. RESULT



Provides the fastest route, accurate ETA and reroutes in real-time.

EXAMPLE



If an accident occurs on the main road, Google Maps instantly finds an alternate route to save time and fuel.

BENEFITS

- Saves travel time
- Reduces fuel consumption
- Avoids traffic & accidents
- Accurate arrival time

2. HEALTHCARE: MEDICAL IMAGING

Detecting diseases early and improving diagnosis using AI.

1. DATA COLLECTION



- X-rays
- MRI scans
- CT scans
- Patient history
- Lab reports

2. MODEL TRAINING



Deep learning models are trained using thousands of healthy and diseased images.

3. DISEASE DETECTION



AI detects patterns and spots abnormalities like tumors, pneumonia and fractures.

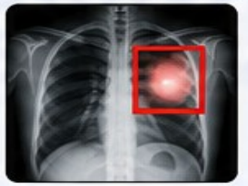
4. RESULT



Assists doctors in faster and more accurate diagnosis and treatment.

EXAMPLE

A chest X-ray is scanned.



AI highlights a suspicious area that may indicate early pneumonia.

BENEFITS

- Early disease detection
- Higher accuracy
- Faster diagnosis
- Better patient outcomes

KEY COMPARISON

Aspect	Google Maps	Healthcare
Goal	Find the best route	Detect diseases early
Data Used	Location, traffic, roads, weather	X-rays, MRI, CT scans, patient data
Technology	Machine Learning, Real-time Analytics	Deep Learning, Computer Vision
Outcome	Faster travel, less congestion	Accurate diagnosis, better treatment
Example	Avoid traffic and reach on time	Detect tumor in medical on time

THE DATA SCIENCE PROCESS (COMMON TO ALL)



Collect Data
Gather data from multiple sources

Explore & Analyze
Find patterns and relationships

Build Models
Train ML/DL models on data

Generate Insights
Make predictions and recommendations

Deliver Value
Better decisions and outcomes

From Data to Decisions – Smarter Navigation, Better Healthcare, Stronger Future.

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DRIVING BUSINESS VALUE THROUGH DATA

How Data Science Creates Impact

1 BETTER DECISIONS



Strategies are based on hard evidence, not human intuition.

Example: A retail store analyzes sales data to understand which products sell more in summer.



Results in higher profits and reduced risks.

2 AUTOMATION



Repetitive tasks, like sorting emails or approving loans, are completely automated.

Example: Banks use AI to automatically review loan applications.



Saves time, reduces errors, and improves efficiency.

3 FASTER PREDICTIONS



Businesses can forecast complex market trends instantly.

Example: A weather forecast system predicts rainfall using past weather data.



Helps in quick planning and staying ahead.

4 COST REDUCTION



Optimizes supply chains and reduces wasted resources.

Example: A logistics company optimizes delivery routes to reduce fuel consumption.



Lowers operational costs and boosts margins.

5 PERSONALIZATION



Customers receive exactly what they want, precisely when they want it.

Example: Netflix recommends movies based on your watch history.



Improves customer satisfaction and loyalty.

6 FRAUD DETECTION



Secures financial systems by spotting unusual behavior anomalies instantly.

Example: Banks detect unusual credit card transactions in real-time.



Prevents fraud and protects customers.

THE DATA SCIENCE PROCESS (Powering All Benefits)



1. Collect Data
Gather data from multiple sources.

2. Prepare Data
Clean, organize and transform data.

3. Analyze Data
Find patterns, trends and insights.

4. Build Models
Train ML/AI models on data.

5. Make Predictions
Generate accurate predictions.

6. Deploy Solutions
Implement solutions in real-world.

7. Monitor & Improve
Continuously monitor and improve results.

Data turns into decisions. Decisions drive value. Value drives growth!

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NAVIGATING THE HURDLES OF DATA IMPLEMENTATION

Key Challenges in Data Science Implementation

1 DATA QUALITY

"Garbage In, Garbage Out."
Models fail if data is inaccurate, biased, or incomplete.

EXAMPLE

Student databace with missing, duplicate, or incorrect values leads to wrong predictions.

Roll No	Attendance	Marks
101	95	85
102	NULL	80
102	NULL	80
103	250	70



SOLUTION

- Clean and validate data
- Remove duplicates
- Handle missing values
- Keep data updated

2 DATA PRIVACY

Handling sensitive user information requires strict security and adherence to laws like GDPR.

EXAMPLE

Hospital stores patient details. If data is leaked, privacy is compromised.



SOLUTION

- Encrypt sensitive data
- Restrict access
- Use secure authentication
- Follow data privacy laws
- Regular security audits

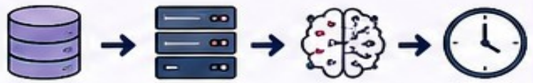


3 COMPLEXITY

Building accurate ML models requires high-end computing power, time, and significant investment.

EXAMPLE

Training a self-driving car model needs millions of images and powerful GPUs.



SOLUTION

- Use cloud computing
- Leverage distributed systems
- Optimize algorithms
- Plan resources efficiently



4 TALENT GAP

Finding professionals with the right mix of skills in data science is difficult.

EXAMPLE

A company needs experts who understand data, algorithms, and business.



SOLUTION

- Train existing employees
- Hire skilled professionals
- Collaborate with universities
- Encourage continuous learning



KEYS TO OVERCOME CHALLENGES



Good Data + Privacy + Technology + Talent = Successful Data Science Implementation

ADITYA UNIVERSITY

THE ASCENDING VALUE OF ADVANCED ANALYTICS

From Understanding the Past to Recommending the Best Future Actions

INCREASING VALUE & COMPLEXITY

1 DESCRIPTIVE ANALYTICS

What happened?



Summarizes historical data to show what has already happened.

EXAMPLE



A company's dashboard shows last month's total sales was ₹10 Lakhs.

Output: Reports, Dashboards, Summaries

2 DIAGNOSTIC ANALYTICS

Why did it happen?



Analyzes data to find the root cause behind an event.

EXAMPLE



Sales dropped because of a website outage that prevented orders.

Output: Insights, Root Causes

3 PREDICTIVE ANALYTICS

What will happen next?



Uses machine learning and statistical models to predict future outcomes.

EXAMPLE



AI predicts next month's sales will be ₹12 Lakhs.

Output: Predictions, Forecasts

4 PRESCRIPTIVE ANALYTICS

What should we do?



Recommends the best actions to achieve desired outcomes.

EXAMPLE



System recommends offering a 10% discount and increasing ads on social media.

Output: Recommendations, Actions

COMPARISON OF ANALYTICS TYPES

Aspect	Descriptive Analytics	Diagnostic Analytics	Predictive Analytics	Prescriptive Analytics
Main Question	What happened?	Why did it happen?	What will happen next?	What should we do?
Purpose	Summarize past data	Find reasons & patterns	Forecast future outcomes	Suggest best actions
Techniques	Reports, Dashboards, Charts	Drill-down, Correlation, Root Cause Analysis	Machine Learning, Regression, Time Series, AI	Optimization, Simulation, AI, Decision Models
Example	Last month sales = ₹10 Lakhs	Sales dropped due to website outage	Next month sales = ₹12 Lakhs	Recommend 10% discount and increase ads
Business Value	★	★★	★★★	★★★★

EXAMPLE: STUDENT PERFORMANCE



Descriptive: 100 students appeared, 80 students passed.

↓



Diagnostic: Students with attendance below 75% mostly failed.

↓



Predictive: AI predicts 15 students are likely to fail in the final exam.

↓



Prescriptive: Recommend remedial classes, counseling & extra study materials.

THE ANALYTICS JOURNEY



Collect Data
Gather raw data from various sources.

→



Process Data
Clean, transform and prepare the data.

→



Analyze Data
Apply appropriate analytics techniques.

→



Generate Insights
Discover patterns, trends and opportunities.

→



Make Better Decisions
Take smart actions to drive business growth.



Right Data → Right Insights → Right Actions → Better Results

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SYNTHESIZING THE FUNDAMENTALS OF DATA SCIENCE

The Building Blocks, Process, Tools, and Real-World Impact

1 THE FOUNDATION

The core disciplines that build Data Science.

MATHEMATICS

$\sqrt{x} \times \mp$

$a^2+b^2=c^2$

Provides formulas and logic to build models and solve problems.

Example: Calculating loan interest, optimizing routes.

STATISTICS

Helps collect, analyze, and interpret data with uncertainty.

Example: Finding that students with >85% attendance score better.

COMPUTER SCIENCE

Gives programming skills and algorithms to process data.

Example: Python reads 50,000 records in seconds.

DOMAIN KNOWLEDGE

Understanding the business or real-world context of data.

Example: A doctor interprets medical reports better than code.

DATA SCIENCE FUNDAMENTALS

Turning Data into Intelligence and Better Decisions

2 THE PROCESS

The step-by-step journey of a Data Science project.

1. **Business Understanding** — Define the problem and objectives. *Example:* Predict students likely to fail.
2. **Data Collection** — Gather data from various sources. *Example:* Attendance, marks, activities from database.
3. **Data Cleaning** — Remove missing, duplicate, and incorrect data. *Example:* Fix wrong marks, remove duplicates.
4. **Data Analysis** — Explore data to find patterns and insights. *Example:* Attendance impacts marks.
5. **Model Building** — Train ML models to make predictions. *Example:* Train model to predict failures.
6. **Deployment** — Deploy model and share insights. *Example:* Dashboard for faculty.

3 THE ECOSYSTEM

Tools and technologies that power Data Science.

Languages	Python, SQL, R
Libraries	Pandas, NumPy, scikit-learn, TensorFlow, PyTorch
Visualization Tools	Tableau, Power BI, Matplotlib
Analytics Types	Descriptive (What happened?) → Diagnostic (Why did it happen?) → Predictive (What will happen?) → Prescriptive (What should we do?)

4 THE VALUE

Transforming data into real-world impact.

- **E-Commerce (Amazon)** — Recommends products based on user behavior.
- **Entertainment (Netflix)** — Suggests movies and shows you may like.
- **Navigation (Google Maps)** — Finds the fastest and safest routes.
- **Healthcare** — Detects diseases using X-rays, MRI, and AI.
- **Banking** — Detects fraud and prevents financial losses.
- **Education** — Predicts student performance and improves learning.
- **Agriculture** — Predicts crop yield and optimizes resources.
- **Manufacturing** — Predicts machine failures and reduces downtime.

THE COMPLETE DATA SCIENCE JOURNEY

Raw Data (From Multiple Sources) → Clean Data (High Quality Data) → Explore & Analyze (Find Patterns & Insights) → Build Models (Machine Learning & AI) → Visualize & Deploy (Dashboards, Reports, APIs) → Better Decisions & Actions (Business Impact)

KEY TAKEAWAY

Data Science combines a strong foundation, a clear process, the right tools, and real-world applications to solve problems, make predictions, and create meaningful value for businesses and society.

★ RIGHT DATA + RIGHT TOOLS + RIGHT PROCESS = SMART DECISIONS AND BETTER FUTURE!

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Testing Your Comprehension of Core Concepts

Question 1: Which of the following is NOT a core phase of the Data Science Lifecycle?

- A) Data Preparation
- B) Exploratory Data Analysis (EDA)
- C) Hardware Manufacturing
- D) Model Deployment

Question 2: Which type of analytics answers the question: "What will happen next?"

- A) Descriptive Analytics
- B) Predictive Analytics
- C) Diagnostic Analytics
- D) Prescriptive Analytics

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Evaluating Industry Applications and Fundamentals

Question 3: Data Science is best described as an intersection of which fields?

A) Physics, Chemistry, and Biology

B) Marketing, Sales, and HR

C) Statistics, Mathematics, Computer Science, and Domain Knowledge

D) Web Design, Networking, and Hardware

Question 4: How does Netflix primarily use Data Science?

A) To manufacture smart TVs

B) To build personalized content recommendation engines

C) To optimize food delivery routes

D) To detect credit card fraud at banks

NotebookLM